

What One Description Conveys to Each Observer.

An Observation Theory Companion to “Rate, Leakage, and Distortion for Gaussian Sources”

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Abstract

A description of data is usually written for one reader and then held by others. Observation Theory treats every party that holds a description as an observer with its own view of the source, its own geometry for weighing errors, and its own budget, and it asks what the same description conveys to each of them. This supplement reads the results of a companion paper on Gaussian sources in that language. The paper’s theorems are stated and proved there in standard information theory, and nothing here adds to them. What this supplement adds is the account of which of those results need the observer to be explicit and what each one says about observers. Four do. What a description conveys to an observer is not fixed by what that observer sees. A cost that is relative to an observer exists only when the decoder’s budget is a single number. The observer’s geometry changes which aspect of the source an optimal description reads, in a way that no theory with a single reader reproduces. And the description that conveys least to an observer at a given rate is the one that also describes what that observer sees. The rest of the paper, including its region and its convexity, is standard and needs no observer.

1 One description, two observers

An encoder sees a source and writes a description for a decoder that must reconstruct part of it. A second party holds the same description together with its own noisy view of the source, and it never decodes. Both parties are observers of one description. The decoder is the observer the description was written for. The second party is an observer the description was not written for, and what the description conveys to it is the information it gains beyond what it already sees.

Standard information theory has the pieces. The rate is what the decoder pays. What the second party gains is the conditional mutual information between the source and the description given its view, the quantity that secure source coding calls the leakage to an eavesdropper. The region of rates and leakages is known in single-letter form, and for Gaussian sources the companion paper [1] works out its structure.

Observation Theory asks a different question of the same objects. It asks how the conveyed content depends on who the observer is. The answer in this setting is that it depends on the observer’s geometry as well as on its data, and the companion paper’s theorems make that dependence exact. This supplement collects the four results where the observer is essential, states what each one says about observers, and marks plainly which parts of the paper owe nothing to the observer at all.

2 The decoder and the second party are both observers

In Observation Theory an observer is a triple $O = (C, G, B)$. The map C is what the observer reads of the source, the matrix G is the geometry in which it weighs errors, and B is its budget [2]. The content a description conveys to an observer that holds side information is the conditional mutual information between the source and the description given that side information, the second coordinate of the rate–content region of [2].

The companion paper has two observers. The decoder reads $C_1 = F$, the linear part $Y = FT$ of the source it must reconstruct, weighs errors with $G_1 = W$, and has the distortion budget D . The second party, called the holder in the paper, reads $C_2 = H$ through the noisy view $S = HT + U$, and the content conveyed to it is the leakage $I(T; \hat{Y} \mid S)$. The encoder sees the whole source T , which may include a context that neither observer’s reconstruction requires. The following table records the correspondence.

Observation Theory	Companion paper
observer the description is written for	decoder, reading $Y = FT$ under weight W
observer the description is not written for	holder, seeing $S = HT + U$
content conveyed to an observer with side information	leakage $I(T; \hat{Y} \mid S)$
budget of the first observer	distortion budget D
geometry of the second observer	information matrix $J = H^\top \Sigma_U^{-1} H$
what the encoder sees	the source T , context included

Every entry on the right is standard. Secure source coding introduced the leakage [4], and the vector Gaussian versions of the problem were treated by Ekrem and Ulukus [5] and by Xu, Chen, Chen, and Jin [6]. Renaming them does not create a result. A result belongs to Observation Theory only if it cannot be stated, or would not have been looked for, without making the second observer explicit. The next four sections are the results of the companion paper that pass that test.

3 What a description conveys to an observer is not fixed by what the observer sees

One might expect the content conveyed to the holder to depend only on the joint law of what the decoder must reconstruct and what the holder sees. It does not. The scalar instances in Section X give two sources with the same joint law of (Y, S) , up to a scaling of S , whose least leakages at the same budget are 1.1880 and 1.2105 bits. The coupled instances give four such sources whose leakage-optimal descriptions reduce the error along 42.83° , 37.61° , 33.75° , and 30.84° at the same budget.

The difference is the encoder’s view. The encoder sees a context that neither observer reconstructs, and the context is correlated with what the holder sees. So the content conveyed to the second observer is a function of three things, the first observer’s task, the second observer’s view, and what the writer of the description sees. A theory that models only the two observers cannot predict it. This is the reason the companion paper’s setting has an encoder that observes more than it describes.

4 A cost relative to an observer exists only when the budget is a single number

Suppose the decoder's requirement is a full error covariance, so that every direction of the source must be reconstructed to a prescribed accuracy. Then Proposition 4 shows that one description is at once the cheapest for the decoder and the one that conveys least to the holder, and there is nothing to trade. The result is implied by the bounds of Ekrem and Ulukus [5].

The tradeoff appears only when the decoder's requirement is a single number, a weighted total distortion, because only then does the encoder choose how to spread the error over directions. In the language of Observation Theory, the first observer's budget must leave freedom in its own geometry before the second observer's geometry can matter. Every other result below lives in that freedom.

5 The observer's geometry changes what an optimal description reads

This is the central finding, and it is the one a theory with a single reader cannot produce.

A description written for a single observer under a weighted quadratic distortion reads fixed directions of the source at every budget. The budget sets how many directions are described and how finely, never which ones (Proposition 19). Principal components, reverse water-filling, and the Gaussian information bottleneck all behave this way.

When a second observer is present, the best one-dimensional description reads a direction that turns as the budget changes, unless that direction is shared by the decoder's geometry and the holder's (Theorem 20). In the coupled instance the leakage-optimal description reduces the error along 47.48° at one budget and 41.15° at another. Corollary 23 turns this into a statement about theories. No problem with a single reader and a fixed weighted quadratic distortion has the leakage-optimal descriptions as its optimal descriptions at two such budgets. The turning is therefore not a feature of a richer single-observer model. It is the trace of the second observer.

The cost of ignoring it is explicit. A description that keeps a fixed direction pays an excess in content conveyed to the holder that Proposition 24 and Theorem 25 bound from below. For a scalar source with a nearly clean context, every fixed direction conveys at least 0.0245 bits more than necessary at one of two budgets, which exceeds the least leakage 0.0216 bits at the second budget.

The turning comes from the misalignment of the two observers' geometries, not from the context. A described variable with no context at all turns whenever the decoder's weight and the holder's information share no eigenvector.

6 To convey little to an observer, describe what that observer sees

The general solution has a reading that is natural in Observation Theory. Theorem 7 shows that every frontier description is reverse water-filling on the decoder's weight tilted by the information the holder's view adds. Corollary 8 restates this. The description that conveys least to the holder at a given rate is the rate-distortion optimal description for a decoder that must reconstruct the holder's own view as well as its target, with a weight that the solution itself fixes.

The reason is an identity. The content conveyed to the holder is the rate minus the information the description carries about what the holder sees. So a description that already tells the holder

much of what it sees has little left to convey beyond it. In the language of the theory, the first observer’s effective geometry is its own geometry plus a copy of the second observer’s view, weighted by how uncertain that view remains after the description.

For a scalar described variable this becomes a single explicit curve (Theorem 18). Every optimal description, at every budget and every weighting of the two costs, reads a direction on the path from the decoder’s own target toward the target corrected by the holder’s information. The rate-optimal description sits at one end, and the description that conveys least sits at a point that moves with the budget.

7 What does not need Observation Theory

Most of the companion paper is standard information theory, and its value there does not depend on the reading above. The single-letter region is due to Villard and Piantanida [4]. Without decoder side information it is a union of the matrix-distortion regions of Ekrem and Ulukus [5], and the paper proves the Gaussian coding theorem directly (Theorem 3). The convexity of both costs in the error covariance (Theorem 5), the dimension bound (Proposition 6), the closed forms in commuting geometry and at low distortion (Theorem 13, Theorem 15), the one-dimensional pencil (Theorem 16), the monotone iteration (Proposition 11), and the treatment of decoder side information (Theorem 30) are results about a convex program. They would be stated the same way by a reader who had never heard of observers.

8 Open questions about several observers

The companion paper has one observer the description is written for and one it is not. Observation Theory asks the same questions of more.

Several holders. With holders that see different views, the content conveyed to each is a separate cost, and the frontier becomes a surface. Whether the tilt of Theorem 7 becomes a sum of tilts, one per holder, and whether the optimal direction can turn toward one holder and away from another, is open.

Observers on both sides. When the decoder also has side information and neither side dominates, the second auxiliary of the secure source coding region can matter. Xu, Chen, Chen, and Jin [6] establish that Gaussian auxiliaries are optimal under a matrix distortion and full-rank views. Whether the observer effects of Sections 5 and 6 survive with a single-number budget and a rank-deficient holder is the main open question.

Observers with rich views. For a holder whose view has rank above one and a described variable of dimension above one, the frontier is the limit of an iteration of closed-form steps (Proposition 11) but has no known closed form.

Beyond Gaussian sources. The non-determination of Section 3 and the turning of Section 5 were found in the Gaussian case. Whether they hold for finite alphabets, where the single-letter region is also known [4], is open.

Where this fits

The observer triple and the rate-content region are defined in [2], which also treats two observers on one description. The vector form of the two-observer results, with several described coordinates, is in [3]. The scripts that check every number quoted here are listed in the companion paper’s appendix on numerical checks.

References

- [1] A. H. Bond, “Rate, leakage, and distortion for Gaussian sources: Water-filling on a tilted weight,” manuscript, 2026.
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